Sample Paper 20

Class - 12th Exam - 2025 - 26

Mathematics (Code-041)

Time : 3 Hours **Max. Marks : 80**

General Instructions :

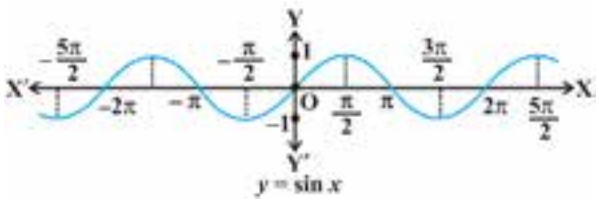
Read the following instructions very carefully and strictly follow them :

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided some questions.
9. Use of calculators is not allowed.

SECTION - A

Section A consists of 20 questions of 1 mark each.

1. What is the maximum value of $y = \sin x$ observed in the graph?



(a) 0

(b) 1

(c) –1

(d) 2

2. Which of the following is a homogeneous differential equation?

(a) $x^2 y\,dx - (x^2 + y^2)\,dy = 0$

(b) $(xy)\,dx - (x^4 + y^4)\,dy = 0$

(c) $(2x + y - 3)\,dy - (x + 2y - 3)\,dx = 0$

(d) $(x - y)\,dy = (x^2 + y + 1)\,dx$

3. A ball thrown vertically upwards according to the formula $s = 13.8t - 4.9t^2$, where s is in metres and t is in seconds. Then its velocity at $t = 1$ sec is

(a) 6m/ sec (b) 4 m/ sec

(c) 2 m/ sec (d) 8 m/ sec

4. The degree of the equation $\left(\frac{d^2y}{dx^2}\right)^2 - x\left(\frac{dy}{dx}\right)^3 = y^3$ is

(a) 0 (b) 1

(c) 2 (d) 3

5. If l, m, n are the direction cosines of a straight line then

(a) $l^2 + m^2 - n^2 = 1$ (b) $l^2 - m^2 + n^2 = 1$

(c) $l^2 - m^2 - n^2 = 1$ (d) $l^2 + m^2 + n^2 = 1$

6. The direction ratios of the line joining the points (x_1, y_1, z_1) and (x_2, y_2, z_2) are

(a) $x_1 + x_2,\ y_1 + y_2,\ z_1 + z_2$

(b) $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$

(c) $\frac{x_1 + x_2}{2},\ \frac{y_1 + y_2}{2},\ \frac{z_1 + z_2}{2}$

(d) $(x_2 - x_1),\ (y_2 - y_1),\ (z_2 - z_1)$

7. $\frac{d}{dx}\left[\lim_{x \to a} \frac{x^5 - a^5}{x - a}\right] =$

(a) $5a^4$ (b) $5x^4$

(c) 1 (d) 0

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8. The direction cosines of the vector $3\hat{i}-4\hat{j}+12\hat{k}$ is

(a) $\frac{3}{13}, \frac{4}{13}, \frac{12}{13}$ (b) $\frac{3}{13}, \frac{-4}{13}, \frac{12}{13}$

(c) $\frac{3}{\sqrt{13}}, \frac{4}{\sqrt{13}}, \frac{12}{\sqrt{13}}$ (d) $\frac{3}{\sqrt{13}}, \frac{-4}{\sqrt{13}}, \frac{12}{\sqrt{13}}$

9. Of all the points of the feasible region, for maximum or minimum of objective functions, the point lies:

(a) inside the feasible region

(b) at the boundary line of the feasible region

(c) vertex point of the boundary of the feasible region

(d) none of the above

10. The maximum value of $y=2x^3-21x^2+36x-20$ is

(a) -128 (b) -126

(c) -120 (d) None of these

11. $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^9 x\, dx = ?$

(a) -1 (b) 0

(c) 1 (d) $\frac{\pi}{2}$

12. What type of a relation is "Less than" in the set of real numbers?

(a) only symmetric

(b) only transitive

(c) only reflexive

(d) equivalence relation

13. $\int x^5\, dx =$ .....

(a) $\frac{x^6}{6}+k$ (b) $\frac{x^5}{5}+k$

(c) $\frac{x^7}{7}+k$ (d) $\frac{x^8}{8}+k$

14. $\tan^{-1}x+\cot^{-1}x = ?$

(a) 0 (b) 1

(c) $\frac{\pi}{2}$ (d) $-\frac{\pi}{2}$

15. Which of the following is the unit matrix of order 3×3?

(a) $\begin{bmatrix}1&0&0\\1&0&0\\1&0&0\end{bmatrix}$ (b) $\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$

(c) $\begin{bmatrix}0&0&1\\0&0&1\\0&0&1\end{bmatrix}$ (d) $\begin{bmatrix}0&1&0\\0&1&0\\0&1&0\end{bmatrix}$

16. In the figure, point A has coordinates:

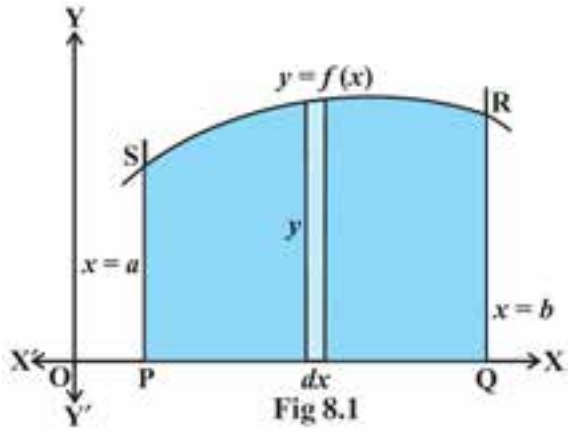
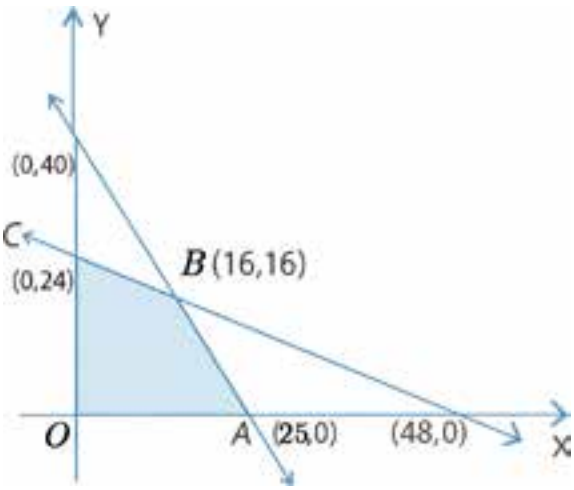


Fig 8.1

(a) (-1,0) (b) (-2/3,0)

(c) (0,0) (d) (1,0)

17. For the given feasible region in the figure, bounded by the lines shown, consider the objective function $z=2x+y$. The feasible region includes the corner points $O(0,0), A(25,0), B(16,16)$, and $C(0,24)$. At the vertex $B(16,16)$, what is the value of z?



(a) 32 (b) 48

(c) 64 (d) 72

18. The radius of a circle is increasing at the rate of 0.4 cm/s. The rate of increase of its circumference is

(a) 0.4π cm/s (b) 0.8π cm/s

(c) 0.8 cm/s (d) None of these

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19. A and B are two events

Assertion : If $P(\overline{A}) = 0.7$, $P(\overline{B}) = 0.5$ and $P(A \cup B) = 0.6$ then $P(A \cap B) = 0.2$

Reason : $P(A \cup B) + P(A \cap B) + P(\overline{A}) + P(\overline{B}) = 2.5$

(a) Both assertion and reason are true and reason is the correct explanation of assertion.

(b) Both assertion and reason are true, but reason is not the correct explanation of assertion.

(c) Assertion is true, but reason is false.

(d) Assertion is false, but reason is true.

20. **Assertion :** If $y = x^3 \cos x$, then $\frac{dy}{dx} = x^3 \sin x + 3x^2 \cos x$

Reason : $\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$.

(a) Both assertion and reason are true and reason is the correct explanation of assertion.

(b) Both assertion and reason are true, but reason is not the correct explanation of assertion.

(c) Assertion is true, but reason is false.

(d) Assertion is false, but reason is true.

SECTION - B

This section comprises of very short answer type-questions (VSA) of 2 marks each.

21. Prove, by Vector method, that the angle inscribed in a semi-circle is a right angle.

22. If $y = \sqrt{x + \sqrt{x + \sqrt{x + ...\text{to}\,\infty}}}$ then $\frac{dy}{dx}$

OR

If $y = \tan(\sin^{-1} x)$ then find $\frac{dy}{dx}$

23. Prove by direction numbers, that the point $(1, -1, 3)$, $(2, -4, 5)$ and $(5, -13, 11)$ are in a straight line.

24. Odds are 8 : 5 against a man, who is 55 years old, living till he is 75 and 4 : 3 against his wife who is now 48, living till she is 68. Find the probability that the couple will be alive 20 years hence.

25. Show that the function $y = ae^{\tan^{-1} x}$ is a solution of the differential equation

$$(1 + x^2)\frac{d^2y}{dx^2} + (2x - 1)\frac{dy}{dx} = 0$$

OR

Show that the function $y = ax + \frac{b}{a}$ is a solution of the differential equation

$$y = x\frac{dy}{dx} + \frac{b}{\frac{dy}{dx}}$$

SECTION - C

This section comprises of short answer type questions (SA) of 3 marks each.

26. If $\vec{a} = \hat{i} - 2\hat{j} - 3\hat{k}$; $\vec{b} = 2\hat{i} + \hat{j} - \hat{k}$ and $\vec{c} = \hat{i} + 3\hat{j} - 2\hat{k}$ then find $\vec{a} \times (\vec{b} \times \vec{c})$.

27. Thus, $\vec{a} \times (\vec{b} \times \vec{c}) = -(\hat{i} + 8\hat{j} - 5\hat{k})$ If $f : R \to R$ is defined by $f(x) = x^2 - 3x + 2$, find $f\{f(x)\}$.

28. If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, then find the value of $A^2 + 3A + 2I$.

OR

Find the values of X and $Y : X + Y = \begin{bmatrix} 7 & 0 \\ 2 & 5 \end{bmatrix}$ and $X - Y = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$.

29. If $y = \log \tan\left(\frac{\pi}{4} + \frac{x}{2}\right)$, show that $\frac{dy}{dx} - \sec x = 0$.

30. The volume of a cube is increasing at a rate of 9 cubic centimetres per second. How fast is the surface area increasing when the length of an edge is 10 cm long?

31. Prove that:

$$\tan\left(\frac{\pi}{4} + \frac{1}{2}\cos^{-1}\frac{a}{b}\right) + \tan\left(\frac{\pi}{4} - \frac{1}{2}\cos^{-1}\frac{a}{b}\right) = \frac{2b}{a}$$

OR

Prove that $4(\cot^{-1} 3 + \text{cosec}^{-1}\sqrt{5}) = \pi$.

SECTION - D

This section comprises of long answer-type questions (LA) of 5 marks each.

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32. Find the particular solution of the differential equation $\frac{dy}{dx} = \frac{xy}{x^2+y^2}$ given that $y = 1$, when $x = 0$.

33. Evaluate $\int \frac{5x+11}{\sqrt{9x^2+25}}\,dx$

OR

Prove that : $\int_0^{\pi/2} \log(\tan\theta + \cot\theta)\,d\theta = \pi \log 2$

34. Solve the following L.P.P. graphically:
Minimise and Maximise

$$Z = 3x + 5y$$

Subject to the constraints:

$$\begin{aligned} 3x - 4y + 12 &\geq 0 \\ 2x - y + 2 &\geq 0 \\ 2x + 3y - 12 &\geq 0 \\ x &\leq 4 \\ y &\geq 2 \\ x &\geq 0 \end{aligned}$$

35. Solve the differential equation :

$$(x-y)\,dy - (x+y)\,dx = 0$$

OR

Solve the differential equation $\frac{dy}{dx} + \frac{y}{x} = \frac{y^2}{x^2}$.

SECTION - E

Case study based questions are compulsory.

36. Pastry is a dough of flour, water and shortening that may be savoury or sweetened. Sweetened pastries are often described as bakers' confectionery. The word "pastries" suggests many kinds of baked products made from ingredients such as flour, sugar, milk, butter, shortening, baking powder, and eggs.



The Sunrise Bakery Pvt Ltd produces three basic pastry mixes A, B and C. In the past the mix of ingredients has shown in the following matrix:

$$\begin{array}{c} \\ \text{Type} \end{array} \begin{array}{c} \begin{array}{ccc} \text{Flour} & \text{Fat} & \text{Sugar} \end{array} \\ \begin{bmatrix} 5 & 1 & 1 \\ 6.5 & 2.5 & 0.5 \\ 4.5 & 3 & 2 \end{bmatrix} \end{array} \begin{array}{l} A \\ B \\ C \end{array}$$

(All quantities in kg)

Due to changes in the consumer's tastes it has been decided to change the mixes using the following amendment matrix:

$$\begin{array}{c} \\ \text{Type} \end{array} \begin{array}{c} \begin{array}{ccc} \text{Flour} & \text{Fat} & \text{Sugar} \end{array} \\ \begin{bmatrix} 0 & 1 & 0 \\ -0.5 & 0.5 & 0.5 \\ 0.5 & 0 & 0 \end{bmatrix} \end{array} \begin{array}{l} A \\ B \\ C \end{array}$$

Using matrix algebra you are required to calculate:

(i) The matrix for the new mix:

(ii) The production requirement to meet an order for 50 units of type A, 30 units of type B and 20 units of type C of the new mix;

(iii) The amount of each type that must be made to totally use up 370 kg of flour, 170 kg of fat and 80 kg of sugar that are at present in the stores.

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37. A steel can, tin can, tin, steel packaging, or can is a container for the distribution or storage of goods, made of thin metal. Many cans require opening by cutting the "end" open; others have removable covers. They can store a broad variety of contents: food, beverages, oil, chemicals, etc.



A tin can manufacturer a cylindrical tin can for a company making sanitizer and disinfector. The tin can is made to hold 3 litres of sanitizer or disinfector.

Based on the above information, answer the following questions.

(i) If r be the radius and h be the height of the cylindrical tin can, find the surface area expressed as a function of r.

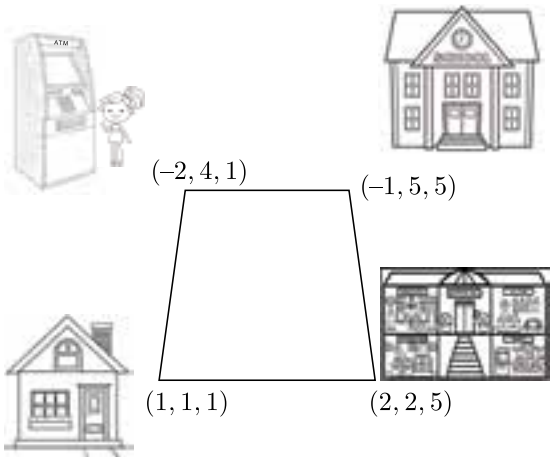
(ii) Find the radius that will minimize the cost of the material to manufacture the tin can.

(iii) Find the height that will minimize the cost of the material to manufacture the tin can.

OR

(iv) If the cost of the material used to manufacture the tin can is 100/m^2 find the minimum cost. $\sqrt[3]{\frac{1500}{\pi}} \approx 7.8$

38. Lavanya starts walking from his house to shopping mall. Instead of going to the mall directly, she first goes to ATM, from there to her daughter's school and then reaches the mall. In the diagram, using co-ordinate geometry the location of each place is given.



Based on the above information, answer the following questions.

(i) What is the distance between House and ATM?

(ii) What is the distance between ATM and school?

(iii) What is the total distance travelled by Lavanya?

OR

(iv) What is the extra distance travelled by Lavanya in reaching the shopping mall?

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